

RAABTA

Script-Aware Multi-Query Reformulation and Evidence Retrieval for Roman-Urdu Questions

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DEVELOPMENT STUDY

The reported measurements use 120 automatically curated and programmatically checked development questions. The 60-question test split remains locked, and independent native-speaker review is pending. This document reports development measurements and does not make a test-set benchmark claim.

Research question

Can script-aware multi-query reformulation improve retrieval effectiveness for noisy Roman-Urdu questions over Urdu-script knowledge bases compared with direct retrieval, single transliteration, and standard hybrid retrieval?

Abstract

Roman Urdu is widely used in informal digital communication but has no standard spelling, is frequently code-switched with English, and differs in script from much authoritative Urdu content. Raabta studies whether a small, controlled set of meaning-preserving query representations can bridge Roman-Urdu questions to Urdu-script evidence. The system preserves the original query, applies conservative normalization, produces supporting-data transliteration and a retrieval-oriented variant, rejects semantic drift, retrieves through BM25 and multilingual E5 routes, fuses rankings with reciprocal rank fusion (RRF), and optionally reranks a bounded candidate set. On 120 development questions, QueryBridge improves Recall@10 from 0.0917 for direct dense retrieval to 0.1667. A depth-20 multilingual reranker improves MRR@10 from 0.0750 to 0.1444, but averages 16.4 seconds on CPU. The application returns exact evidence sentences, sources, traces, latency, and explicit abstention. Results remain provisional because independent language review is pending and the frozen 60-question test split has not been used.

1. Introduction

Roman Urdu creates an information-access mismatch: users often type Urdu using Latin characters, while a large body of relevant knowledge is written in Urdu script. Variation such as *kia/kya*, *hai/hy*, abbreviations, phonetic spelling, named entities, and English insertions weakens exact lexical matching. Direct multilingual dense retrieval can cross scripts, but it may not reliably handle extreme noise or surface-form mismatch. Raabta treats this as a retrieval and evidence problem rather than as a generic chatbot problem.

1.1 Positioning and contribution

Butt, Varanasi, and Neumann (2025) introduced a large-scale Roman-Urdu information-retrieval dataset and multilingual baseline [1]. Raabta does not claim that Roman-Urdu IR is unstudied. Its narrower contribution is to evaluate transparent, controlled multi-query representations over a fixed Urdu-script corpus, combine complementary retrieval routes, measure which components contribute, and expose an evidence-first local application. The research contribution is therefore the script-aware query bridge and its controlled evaluation, not the use of a large language model.

2. Data and frozen protocol

The primary corpus is a deterministic 4,000-article subset of the 20231101 Urdu Wikipedia snapshot [2]. Cleaning preserves article identity, title, URL, raw text, normalized text, and an auditable six-domain assignment. The default passage collection contains 16,352 passages of up to 150 whitespace tokens with 30-token overlap. The passage file SHA-256 is 47648cf679facb9a576841542289767f854c1a27aca6cb8e14e3f3eb1a2e5671. A supporting Roman-Urdu parallel slice supplies transliteration evidence only; it is not used as relevance supervision.

2.1 Diagnostic set

The Raabta Diagnostic Set contains 180 automatically curated title-definition questions balanced across culture, general, geography, history, Pakistan, and science. It includes clean, informal, highly noisy, abbreviated, code-switched, named-entity, short, and slightly ambiguous queries. Passage IDs and exact evidence strings were programmatically checked against the frozen corpus. The split was frozen before tuning: 120 development questions and 60 test questions. Independent native-speaker review is required before making final benchmark claims.

3. System method

QueryBridge creates no more than four traceable candidates: (1) the original query, (2) conservatively normalized Roman Urdu, (3) Urdu-script transliteration derived from a supporting lexicon, and (4) a controlled retrieval-oriented representation. Each candidate records its method, source query, semantic similarity, and accept/reject reason. A pinned multilingual E5-small encoder [3] filters semantic drift at a 0.55 cosine threshold. Accepted variants independently traverse Unicode BM25 and exact normalized dense retrieval. Equal-weight RRF combines incomparable route scores without pretending they are calibrated probabilities.

3.1 Reranking and grounded answers

The top 20 fused candidates can be reranked with Alibaba-NLP/gte-multilingual-reranker-base using pinned model and custom-code revisions. Because the depth-20 cross-encoder is slow on CPU, it is used for offline evaluation; the interactive API currently uses fused retrieval without reranking. The answer layer splits the top passages into sentences, compares accepted QueryBridge representations with source sentences, and returns the strongest coherent one- or two-sentence extract. If no evidence exists or similarity falls below 0.70, it returns an Urdu abstention with no citation.

4. Experimental design

The frozen baselines are direct dense retrieval, one-best deterministic transliteration plus BM25, and a standard raw-query BM25+dense RRF hybrid. Metrics are Recall@1, Recall@5, Recall@10, MRR@10, and nDCG@10. All development metrics are macro averages over 120 questions. Latency excludes cold model loading unless explicitly stated. Robustness is reported by eight query categories. Retrieval-component ablations remove normalization, transliteration, expansion, BM25, dense retrieval, or RRF; the paired before/after reranker measurement isolates reranking.

5. Development results

Table 1 compares all frozen systems. QueryBridge produces the best Recall@10 before reranking, while the cross-encoder primarily improves early ordering. The result supports the development hypothesis but does not establish final generalization.

System	R@1	R@5	R@10	MRR@10	nDCG@10	Mean ms
Direct dense	0.0500	0.0833	0.0917	0.0615	0.0686	51.8
Single transliteration + BM25	0.0000	0.0167	0.0250	0.0074	0.0116	60.8
Standard hybrid	0.0250	0.0667	0.0917	0.0456	0.0565	94.5
QueryBridge + RRF	0.0500	0.1000	0.1667	0.0750	0.0959	444.0
QueryBridge + reranker	0.1250	0.1750	0.1833	0.1444	0.1540	16947.6

Table 1. Development-only retrieval results (n=120; test queries used=0).

Recall@10 by system - development only

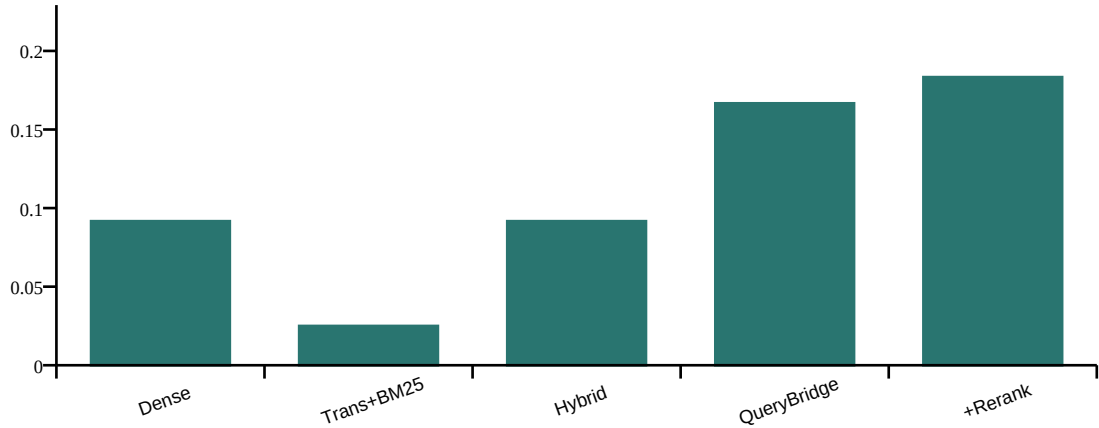


Figure 1. QueryBridge raises candidate recall; reranking gives a smaller additional Recall@10 gain.

5.1 Robustness

Performance varies sharply by query type. Clean queries reach Recall@10 of 0.5000, while abbreviated queries reach 0.0000. Highly noisy, short, and code-switched queries retain some signal, but informal spelling and named entities remain difficult. These small group counts are diagnostic, not population estimates.

Query type	n	Reranked R@10	Reranked MRR@10
abbreviated roman urdu	15	0.0000	0.0000
clean roman urdu	16	0.5000	0.3385
highly noisy roman urdu	15	0.2667	0.2222
informal spelling	16	0.0625	0.0625
named entity	14	0.0714	0.0238
short query	15	0.2667	0.2167
slightly ambiguous	15	0.0667	0.0667
urdu english code switching	14	0.2143	0.2143

Table 2. Reranked robustness by frozen query category.

5.2 Ablation study

Removing transliteration, dense retrieval, or RRF causes the largest Recall@10 losses. BM25 and normalization add smaller positive gains. The expansion rule is not uniformly beneficial: removing it lowers Recall@10 slightly but improves MRR@10. This negative result argues for redesign rather than post-hoc defense. Retrieval controls are measured before reranking to isolate retrieval behavior; a final reranker-retained matrix remains pending for the main PC.

Configuration	R@10	MRR@10	Mean ms	Delta R@10
full no reranker	0.1667	0.0750	425.2	+0.0000
no bm25	0.1417	0.0529	271.1	-0.0250
no dense	0.0833	0.0488	254.8	-0.0833

Configuration	R@10	MRR@10	Mean ms	Delta R@10
no expansion	0.1583	0.0843	354.4	-0.0083
no fusion	0.0583	0.0232	489.2	-0.1083
no normalization	0.1500	0.0715	402.9	-0.0167
no transliteration	0.0667	0.0253	335.8	-0.1000

Table 3. Retrieval-stage leave-one-component-out controls (n=120).

5.3 Error analysis

A traceable 30-case audit balances six observed categories: Roman spelling mismatch, excessive spelling noise, code-switching failure, named-entity mismatch, short/ambiguous query, and irrelevant retrieval. Each row records the real query, gold passage, top retrieved passage and title, affected system, likely cause, and a possible improvement. Labels are deterministic first-pass assignments and explicitly require human review. The dominant engineering remedies are character-level normalization, language-aware token handling, entity aliases, clarification for underspecified queries, and reviewed hard negatives.

5.4 Latency and resource trade-offs

The 16,352 x 384 float32 embedding matrix occupies 25.1 MB and required 42.4 minutes for cold CPU embedding generation and serialization. Direct dense retrieval averages 51.8 ms; QueryBridge averages 444.0 ms. Depth-20 reranking averages 16.4 seconds and observed process resident memory reached 2.49 GB. The quality gain is meaningful offline but impractical for the interactive CPU path at depth 20.

6. Application architecture

The complete local application uses a FastAPI backend and responsive React/Vite frontend. The backend exposes health, query, comparison, source, and configuration endpoints with Pydantic validation and an explicit local CORS allowlist. A query response contains the original query, accepted and rejected variants, exact answer, evidence spans, sources, retrieval trace, non-probabilistic scores, and component latency. Research Mode compares Direct Dense, Single Transliteration, Standard Hybrid, and Raabta. It reports NOT PROVIDED when a free-form query has no gold label and NOT RETRIEVED when supplied verified evidence is absent.

6.1 Evidence and abstention

Every supported answer is a direct substring of a retrieved source passage. The real-corpus smoke query 'pakistan ka capital kya hai' retrieves the Urdu Wikipedia article about Pakistan's capitals at rank 1 and returns the exact sentence stating that Islamabad has been the national or federal capital since 1960. When candidates are absent or evidence similarity is below threshold, the system returns a fixed Urdu abstention, no evidence, and no source. This prevents unsupported generation by construction.

7. Reproducibility

All Python runtime, development, build, and frontend dependencies are pinned. Dataset and model identifiers and revisions are stored in configuration and manifests. The passage and embedding checksums are verified by a portability audit. Seven notebooks execute from top to bottom with fixed seed 20250816 and visible outputs. The main Phase 6 command regenerates tables, figures, latency reporting, and the error audit from measured artifacts. A fresh Python 3.12 environment successfully builds and imports the local package. The project excludes .venv and node_modules so they can be recreated on another PC.

8. Limitations, ethics, and threats to validity

The most important limitation is annotation independence: the initial questions and evidence checks were automated, so a native Roman-Urdu/Urdu reviewer must confirm naturalness, intent, evidence, and failure categories. Title-definition construction may favor encyclopedic queries and the 4,000-article sample is not representative of all Urdu information needs. The semantic drift threshold and deterministic expansion were tuned only for engineering feasibility. Group sizes are small. The test set remains locked. Wikipedia licensing and attribution must be preserved, and the diagnostic CSV should not be published without a redistribution review. The system should be presented as an information-retrieval prototype, not an authoritative answer engine.

9. Conclusion

Raabta demonstrates a complete CPU-first path from noisy Roman-Urdu input to traceable Urdu evidence. On the provisional development split, controlled multi-query retrieval improves Recall@10 over direct dense and standard hybrid baselines, while reranking improves early ranking at substantial latency cost. Transliteration, dense retrieval, and RRF are the strongest observed components; the current expansion rule is mixed. The strongest practical contribution is a transparent system that exposes what was changed, which routes contributed, which evidence supports the answer, and when the system abstains. Final scientific claims await independent review and one-time locked test evaluation.

References

- [1] Muhammad Umer Tariq Butt, Stalin Varanasi, and Guenter Neumann. Roman Urdu as a Low-Resource Language: Building the First IR Dataset and Baseline. LowResNLP 2025, pp. 82-87. <https://aclanthology.org/2025.lowresnlp-1.9/>
- [2] Wikimedia. Wikipedia dataset, Urdu configuration 20231101.ur, frozen revision 3e1f92c331f318af862b87e2319ed5dc26d80f5d. CC BY-SA 3.0 and GFDL. <https://huggingface.co/datasets/wikimedia/wikipedia>
- [3] Liang Wang et al. Multilingual E5 Text Embeddings: A Technical Report. arXiv:2402.05672, 2024. Model card: <https://huggingface.co/intfloat/multilingual-e5-small>
- [4] Muhammad Umer Tariq Butt, Stalin Varanasi, and Guenter Neumann. Low-Resource Transliteration for Roman-Urdu and Urdu. LoResMT 2025, pp. 144-153. <https://aclanthology.org/2025.loresmt-1.13/>

Appendix A. Frozen identifiers

Dense model: intfloat/multilingual-e5-small @ d1d99a1efae6779390caba937d92c54b5bc70e51. Reranker weights: Alibaba-NLP/gte-multilingual-reranker-base @ a6258e9d2b1a11aa7bccdff9efde562bbca4393d. Reranker custom code: 40ced75c3017eb27626c9d4ea981bde21a2662f4. Urdu Wikipedia revision: 3e1f92c331f318af862b87e2319ed5dc26d80f5d. Diagnostic SHA-256: 98f723f1e5cb8f1ece24a11192052e7a2119629703e001084c0bdcadbfe37cfd.

Appendix B. Definition of finalization

Finalization requires: (1) independent native-speaker review of all diagnostic questions and failure labels; (2) corrections made without examining test retrieval outputs; (3) final configuration frozen; (4) one-time evaluation on the 60 locked test questions; (5) replacement of provisional tables and figures; and (6) retention of development and test results as separate, clearly labeled analyses.